

```
In [1]: import pandas as pd
```

```
In [2]: df = pd.read_csv("fra_perfumes.csv")
df.head()
```

Out[2]:

	Name	Gender	Rating Value	Rating Count	Main Accords	Perfumers	Description	
0	9am Afnanfor women	for women	3.73	174	['citrus', 'musky', 'woody', 'aromatic', 'warm...]	[]	9ambyAfnanis a fragrance for women. Top notes ...	https://www
1	9am Dive Afnanfor women and men	for women and men	4.29	842	['fruity', 'woody', 'green', 'warm spicy', 'ar...]	[]	9am DivebyAfnanis a Aromatic Aquatic fragrance...	https://www
2	9am pour Femme Afnanfor women	for women	4.00	68	['fruity', 'musky', 'amber', 'citrus', 'powder...]	[]	9am pour FemmebyAfnanis a Amber fragrance for ...	https://www
3	9pm Afnanfor men	for men	4.50	6,865	['vanilla', 'amber', 'warm spicy', 'cinnamon',...]	[]	9pmbyAfnanis a Amber Vanilla fragrance for men...	https://www
4	9pm pour Femme Afnanfor women	for women	3.49	63	['woody', 'aromatic', 'rose', 'fruity', 'powde...]	[]	9pm pour FemmebyAfnanis a Amber Floral fragran...	https://www

```
In [3]: df.info()
df.isnull().sum()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 70103 entries, 0 to 70102
Data columns (total 8 columns):
#   Column          Non-Null Count  Dtype
---  ---
0   Name             70100 non-null  object
1   Gender           70100 non-null  object
2   Rating Value     63922 non-null  float64
3   Rating Count     63922 non-null  object
4   Main Accords      70103 non-null  object
5   Perfumers        70103 non-null  object
6   Description      70100 non-null  object
7   url              70103 non-null  object
dtypes: float64(1), object(7)
memory usage: 4.3+ MB
```

```
Out[3]: Name             3
        Gender           3
        Rating Value     6181
        Rating Count     6181
        Main Accords      0
        Perfumers        0
        Description      3
        url              0
        dtype: int64
```

```
In [8]: df = df[['Name', 'Gender', 'Main Accords']].dropna()

df['features'] = df['Main Accords'] + " " + df['Gender']

from sklearn.feature_extraction.text import TfidfVectorizer

tfidf = TfidfVectorizer(max_features=3000, stop_words='english')

vectors = tfidf.fit_transform(df['features'])
```

```
In [10]: def recommend_perfume(query):
        query_vec = tfidf.transform([query])
        scores = cosine_similarity(query_vec, vectors)

        top_indexes = scores[0].argsort()[-5:][::-1]

        return df.iloc[top_indexes][['Name', 'Main Accords']]
```

```
In [12]: import pickle

pickle.dump(tfidf, open("tfidf.pkl", "wb"))
pickle.dump(vectors, open("vectors.pkl", "wb"))
pickle.dump(df, open("data.pkl", "wb"))
```

```
In [13]: import pickle

# TF-IDF model save
with open("tfidf.pkl", "wb") as f:
    pickle.dump(tfidf, f)
```

```
# Vectors save
with open("vectors.pkl", "wb") as f:
    pickle.dump(vectors, f)

# DataFrame save
with open("data.pkl", "wb") as f:
    pickle.dump(df, f)

print("✅ All files saved successfully!")
```

✅ All files saved successfully!

In []: